

Traffic Collision Data Analysis Report

1. Introduction

This project analyzes large-scale California traffic collision data using PySpark to uncover spatial, temporal, and environmental patterns affecting road safety. Due to the large size of the dataset, PySpark was used for scalable processing, while Pandas and visualization libraries were used for exploratory analysis on sampled data.

2. Data Preparation

The dataset was loaded from CSV/SQLite files into PySpark DataFrames. Schema inspection ensured correct data types and structure. Sampling was applied only during visualization to maintain system stability while preserving representative trends.

3. Data Cleaning

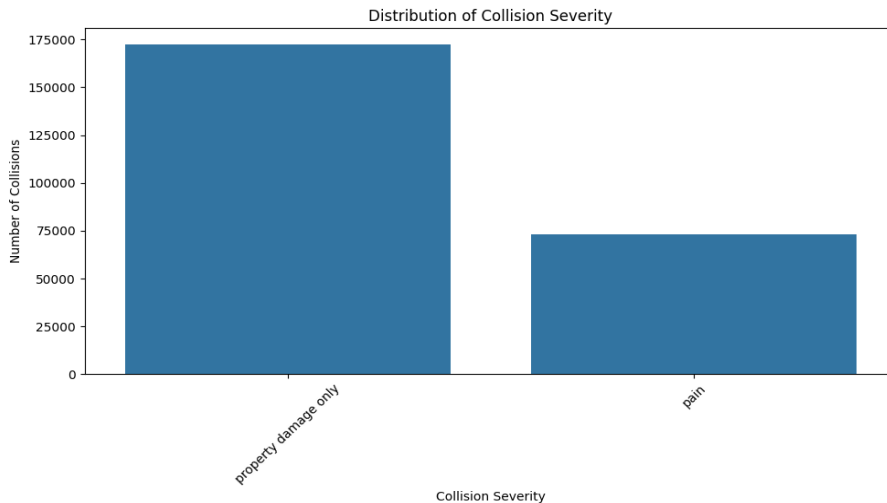
Missing values were examined column-wise across all tables. A rule-based approach was used instead of statistical imputation. Numeric count fields (killed_victims, injured_victims) were killed with zero, categorical variables were killed with 'Unknown', and records with missing critical attributes were removed. Duplicate records were eliminated. Outliers were identified using the Interquartile Range (IQR) method but retained to preserve rare but severe events.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted using univariate, segmented univariate, and bivariate techniques. Visualizations were generated using Pandas after sampling from Spark DataFrames.

4.1 Collision Severity Distribution

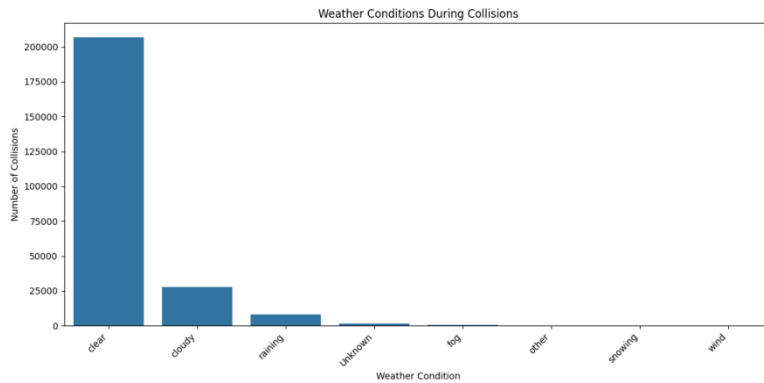
Visualisation: Bar chart of collision severity.



- **Analysis:** Univariate analysis. Result: Most collisions are low severity, while fatal collisions are relatively rare.
- **Insight:** High-frequency low-severity crashes dominate overall statistics but do not reflect true societal impact.
- **Outcome:** Road safety policies should prioritise reducing high-severity and fatal collisions rather than total counts alone.

4.2 Weather Conditions During Collisions

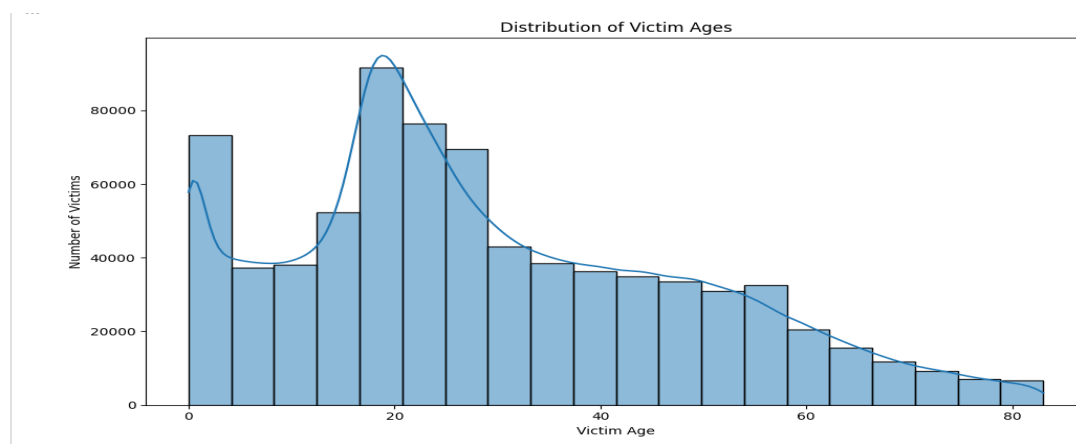
Visualisation: Bar chart of weather conditions.



- **Analysis:** Segmented univariate analysis. Collision counts were analysed across different weather conditions using categorical aggregation Result: Most collisions occur during clear weather.
- **Insight:** Traffic exposure and driving volume influence collisions more than adverse weather alone.
- **Outcome:** Road safety planning should focus on normal driving conditions.

4.3 Victim Age Distribution

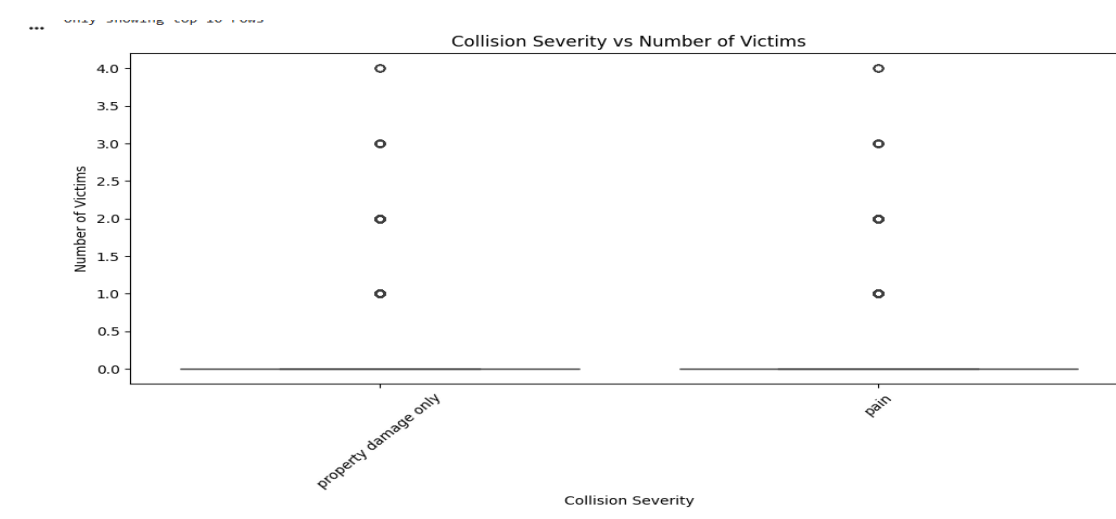
Visualisation: Histogram of victim ages.



- **Analysis:** Univariate analysis. Result: Young and middle-aged adults are most affected.
- **Insight:** Active working population faces higher road risk due to frequent travel. **Outcome:** Targeted awareness campaigns for commuters and working-age drivers.

4.4 Collision Severity vs Number of Victims

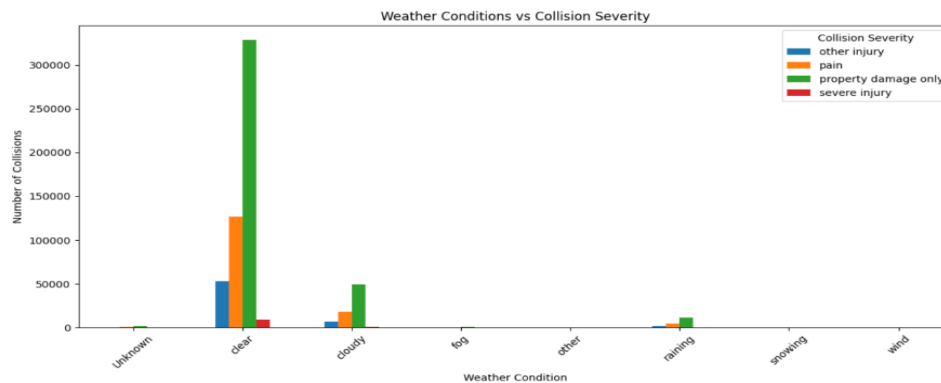
Visualisation: Scatter/box plot.



- **Analysis:** Bivariate analysis. Result: Higher severity collisions is associated with an increased number of victims.
- **Insight:** Multi-victim crashes tend to escalate injury severity and emergency impact.
- **Outcome:** Improved emergency response planning should prioritise rapid intervention for multi-victim collisions.

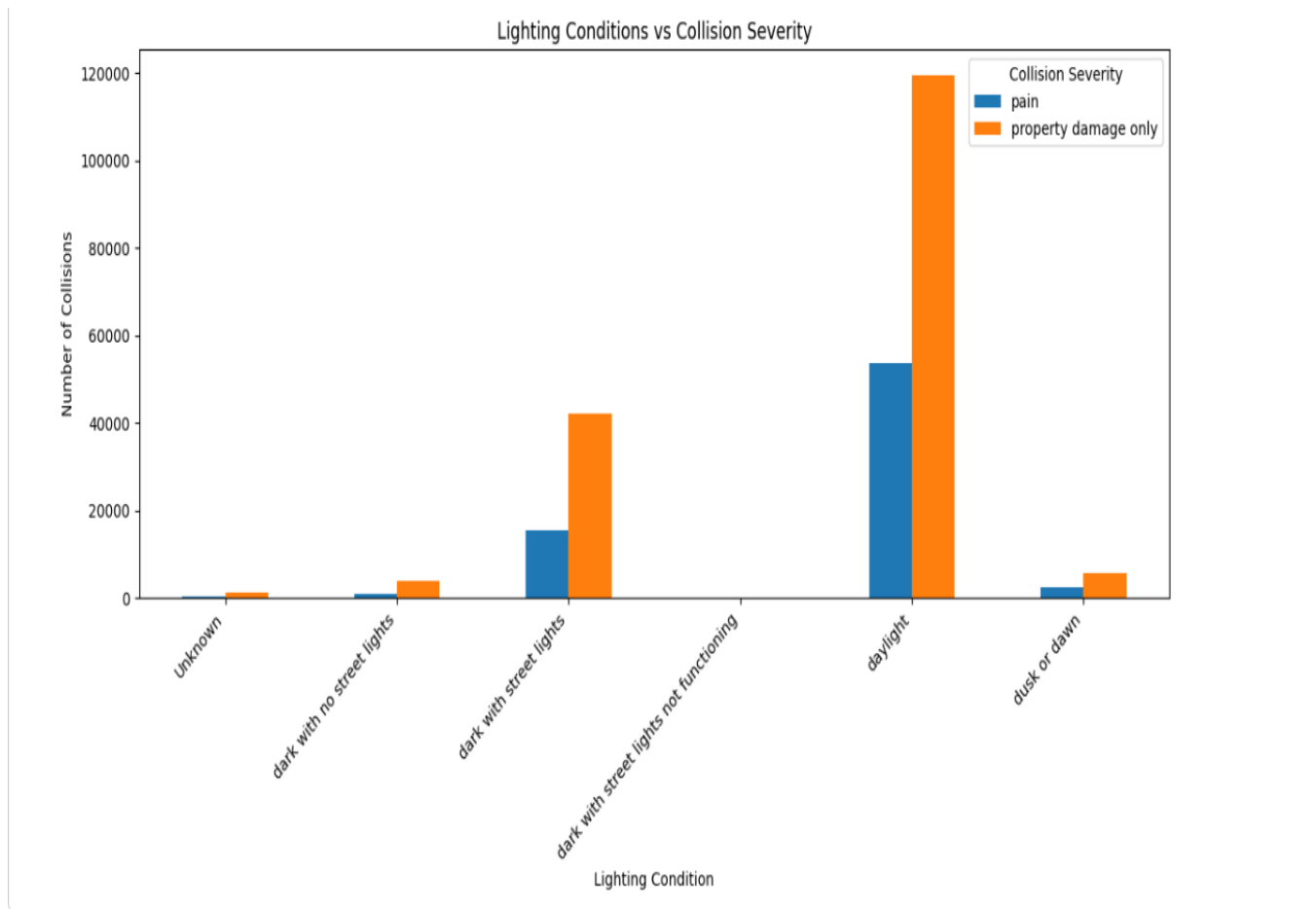
4.5 Weather Conditions vs Collision Severity

Visualisation: Bar plot.



- **Analysis:** Bivariate analysis
- **Result:** Most severe and fatal collisions occur during clear weather rather than adverse weather conditions.
- **Insight:** Higher traffic volume and speed during clear weather increase collision severity despite favorable driving conditions.
- **Outcome:** Road safety interventions should target normal weather conditions through speed control and awareness programs.

4.6 Lighting Conditons vs Collision Severity



- **Analysis:** Bivariate categorical analysis. Result: Poor or no lighting correlates with higher severity.
- **Insight:** Visibility is a major risk factor. Reduced visibility significantly increases the risk of serious injuries.
- **Outcome:** Improve street lighting in high-risk areas can reduce severe collision outcomes.

4.7 Weekday-Wise Collision Trends

Visualization: Bar charts (weekday, monthly, hourly).

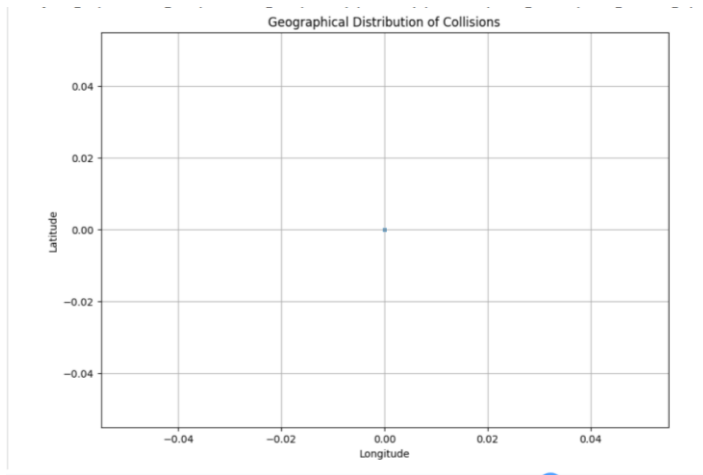
Analysis: Spatial analysis. Collision counts were aggregated and analyzed spatially at the county level.

Result: Collisions cluster in urban counties. A small number of urban counties account for a disproportionately high number of collisions.

Insight: Risk is geographically concentrated. Urban density and traffic concentration are key contributors to collision frequency.

Outcome: Hotspot-based infrastructure investment.

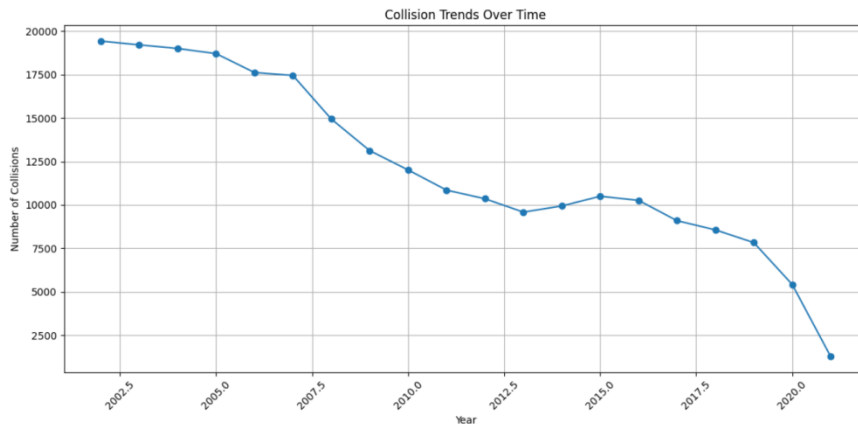
4.9 Collision Analysis by Geography



4.9 Collision Trends Over Time

Yearly Trend

Visualization: Line bar



Analysis: Segmented time-series analysis. Year-wise aggregation of collision counts was conducted to observe long-term trends.

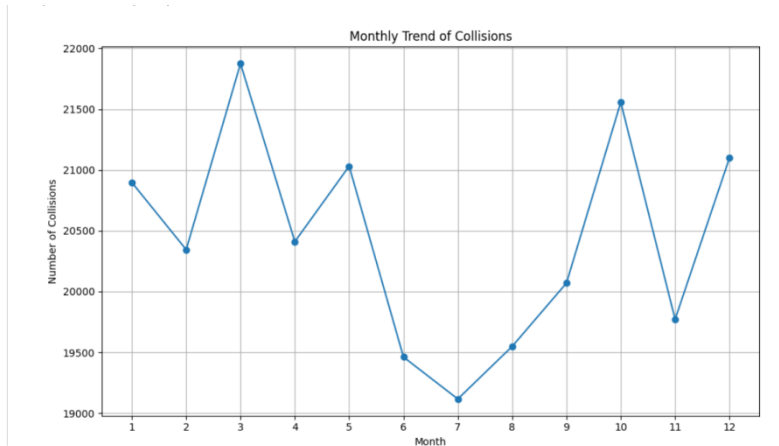
Result: Collision volumes show year-to-year variation with identifiable peak years.

Insight: Changes in traffic volume, population growth, and mobility patterns influence annual collision trends.

Outcome: Long-term safety planning should consider evolving traffic demand and urban expansion.

Monthly Trend

Visualization: Line bar



Analysis: Segmented time-series analysis. Monthly collision counts were analysed to detect seasonal patterns.

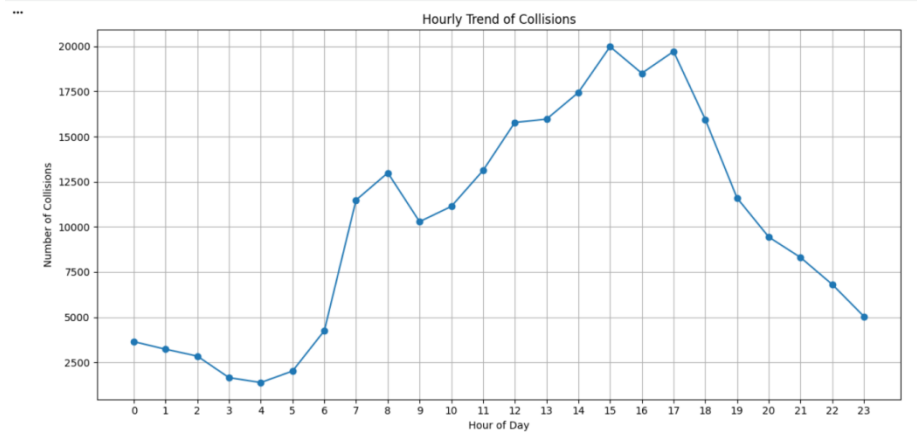
Result: Certain months consistently show higher collision frequencies.

Insight: Seasonal travel patterns and holiday-related mobility contribute to collision fluctuations.

Outcome: Season-specific traffic management and enforcement strategies can reduce risk.

Weekly Trend

Visualization: Line bar



Analysis: Segmented time-series analysis. Hourly collision distribution was analysed using time-of-day aggregation.

Result: Collision peaks occur during morning and evening commute hours.

Insight: Congestion, time pressure, and driver fatigue increase collision likelihood during peak hours.

Outcome: Targeted enforcement and optimised signal timing during peak hours can improve road safety.

5. ETL Query Results

ETL operations were implemented using PySpark SQL. Results include identification of top counties, peak months, common weather conditions, fatal collision percentage, dangerous hours, road surface and lighting conditions contributing to collisions.

a) Recommendations to improve road safety

- The Identifying high-risk locations and peak accident times to guide infrastructure improvements and targeted enforcement
- Prioritize high-collision counties and urban corridors for infrastructure upgrades
- Deploy targeted enforcement during peak accident hours (morning and evening commutes)
- Improve intersection design, lane markings, and signage in identified high-risk zones
- Use historical collision data to rank locations by risk rather than spreading resources uniformly

b) Traffic Management Optimization

- Adjust traffic signal timing during peak weekday hours to reduce congestion-related collisions
- Improve road lighting in high-severity zones, especially where night-time crashes are frequent
- Incorporate seasonal and monthly trends into traffic planning (holidays, travel peaks). Design roads assuming collisions occur mostly under normal driving conditions, not just adverse weather

c) Policy Changes for Pedestrian and Cyclist Safety

- Expand dedicated pedestrian crossings and protected bicycle lanes in dense urban areas
- Enforce lower speed limits in zones with high pedestrian and cyclist collision density
- Improve visibility measures (reflective signs, better lighting) near schools, markets, and transit hubs
- Launch behavioral awareness campaigns targeting both drivers and vulnerable road users

d) Identification of High-Risk Zones

- Use county-level and latitude–longitude density analysis to identify collision hotspots
- Classify zones into high, medium, and low risk for proactive intervention
- Focus preventive measures on areas with persistent historical collision patterns, not just recent spikes
- Support local authorities with data-backed hotspot maps for decision-making

e) Impact of Environmental Factors

- Weather alone does not drive collision frequency; traffic volume is the dominant factor
- Poor lighting conditions are strongly associated with higher collision severity

- Road surface conditions reflect exposure rather than inherent danger, but maintenance is still critical
- Environmental factors should be treated as risk multipliers, not sole causes

f) Predictive Modeling and Proactive Safety

Develop predictive models using:

- Time (hour, weekday, month)
- Location (county, coordinates)
- Environmental conditions (lighting, weather)
- Collision severity history
- Use models to anticipate collision hotspots before incidents occur.
- Integrate predictions into traffic policing schedules and infrastructure planning.
- Enable data-driven, preventive road safety strategies rather than reactive responses

Univariate, Segmented & Bivariate Analysis

- Univariate analysis identified dominant patterns (severity distribution, age groups, weather).
- Segmented univariate analysis revealed differences across weekdays, months, and lighting conditions.

- Bivariate analysis showed how severity escalates with victim count and poor visibility.
- Together, these analyses explain where, when, and why collisions occur.

Business and Policy Implications

- Data-driven safety planning reduces economic loss, healthcare burden, and fatalities.
- Targeted interventions are more cost-effective than blanket policies.
- Urban planning informed by collision data leads to safer, more resilient transport systems.

6. Final Insights and Outcomes

The analysis demonstrates strong spatial and temporal clustering of traffic collisions in California. Recommendations include targeted enforcement during peak hours, infrastructure improvements in high-risk zones, enhanced pedestrian and cyclist safety measures, and predictive modeling for proactive intervention. These insights support data-driven traffic safety policy and urban planning.